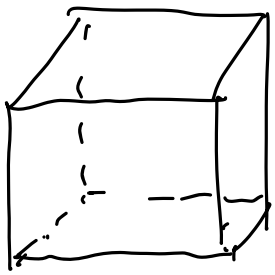
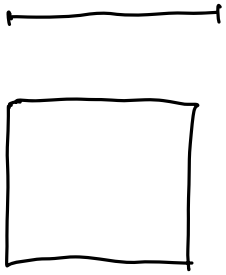
数理方程

第六周

20260403

正交曲线坐标系

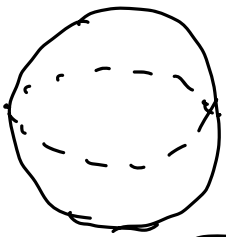




直角坐标系

煮鸡蛋：20°C蛋放进100°C水，多久后中心达到80°C？

近似为球形

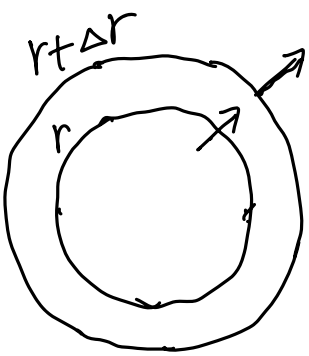


$$u_t = k(u_{xx}+u_{yy}+u_{zz})$$

$$u|_{r=a} = 100°C$$

$u = u(r,t)$

$$u_t = k u_{rr}$$



考虑薄层 $[r, r+\Delta r]$，体积 $4\pi r^2 \Delta r$

r：净流入热量：$4\pi r^2\left(-k\frac{\partial u}{\partial r}\right)\Big|_r$

$r+\Delta r$：净流出：$4\pi r^2\left(-k\frac{\partial u}{\partial r}\right)\Big|_{r+\Delta r}$

$$4\pi r^2 \Delta r\, c\rho \frac{\partial u}{\partial t} = -k\,4\pi r^2 \frac{\partial u}{\partial r}\Big|_r + k\,4\pi r^2 \frac{\partial u}{\partial r}\Big|_{r+\Delta r}$$

热容　密度

$$= 4\pi k \frac{\partial}{\partial r}\left(r^2 \frac{\partial u}{\partial r}\right)\Delta r$$

$\nabla^2 u$ 的径向分量

$$\Rightarrow u_t = \frac{k}{c\rho}\cdot\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial u}{\partial r}\right) = k\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial u}{\partial r}\right)$$

$\frac{k}{c\rho} = k$

$$= k u_{rr} + 2k\frac{2}{r}u_r$$

不是 u_{rr}！坐标系不均匀会产生额外因子

正交曲线坐标系

1) 微分算子怎么写

2) 分离变量

一般几何形状 PDE 用直角坐标处理不便.

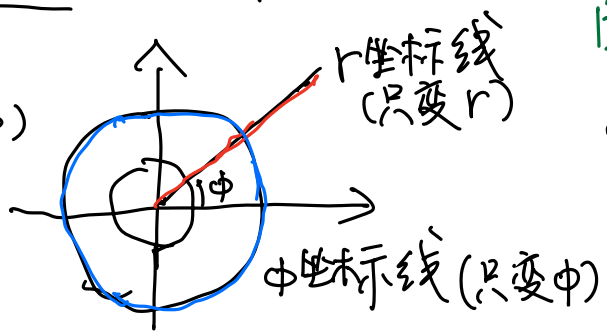
例: 球面 BC $u|_{r=a}=0$

$u=X(x)Y(y)Z(z) \Longrightarrow$?

正交曲线坐标系: 空间任意一点, 所有坐标线互相垂直

固定其他所有坐标值, 只变一个坐标值

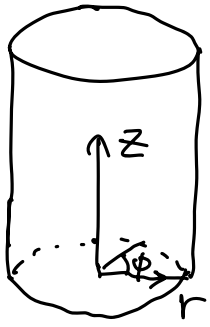
极坐标 (r,ϕ)



$x=r\cos\phi$

$y=r\sin\phi$

柱坐标 (r,ϕ,z)

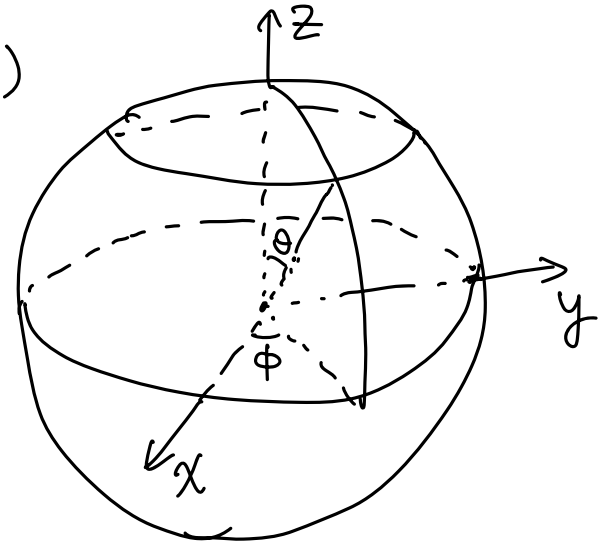


$x=r\cos\phi$

$y=r\sin\phi$

$z=z$

球坐标 (r,θ,ϕ)



$x=r\sin\theta\cos\phi$

$y=r\sin\theta\sin\phi$

$z=r\cos\theta$

抛物坐标 (ξ, η, ϕ)

$x=\sqrt{\xi\eta}\cos\phi$, $y=\sqrt{\xi\eta}\sin\phi$, $z=\frac{1}{2}(\xi-\eta)$

$\xi=r+z\geq 0$, $\eta=r-z\geq 0$, $\phi=\arctan\frac{y}{x}$

$r=\sqrt{x^2+y^2+z^2}=\frac{1}{2}(\xi+\eta)$

$\xi\eta=x^2+y^2=r^2-z^2$
$=(r+z)(r-z)$
$\xi-\eta=2z$
$\Downarrow$
$\xi=r+z$, $\eta=r-z$

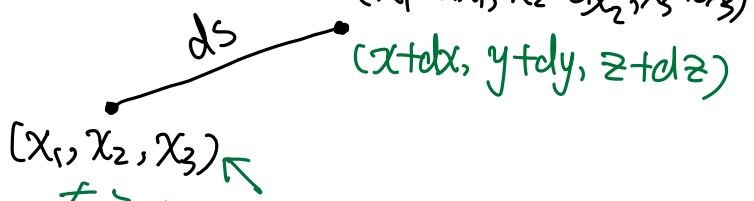
正交曲线坐标系

判别法：度规法

曲线坐标系 (x_1, x_2, x_3)

距离微元

$ds^2=dx^2+dy^2+dz^2$



直角坐标系坐标为 (x, y, z)

$x=x(x_1, x_2, x_3)$
$y=y(x_1, x_2, x_3)$
$z=\cdots$

$=\left(\frac{\partial x}{\partial x_1}dx_1+\frac{\partial x}{\partial x_2}dx_2+\frac{\partial x}{\partial x_3}dx_3\right)^2+\left(\frac{\partial y}{\partial x_1}dx_1+\frac{\partial y}{\partial x_2}dx_2+\frac{\partial y}{\partial x_3}dx_3\right)^2$
$+\left(\frac{\partial z}{\partial x_1}dx_1+\cdots\right)^2$

$=\sum_{i,j=1,2,3} g_{ij}\,dx_i\,dx_j = g_{11}(dx_1)^2+g_{22}(dx_2)^2+g_{33}(dx_3)^2$
$+2g_{12}dx_1dx_2+2g_{13}dx_1dx_3+2g_{23}dx_2dx_3$

度规矩阵

$g_{ij}=g_{ji}=\frac{\partial x}{\partial x_i}\frac{\partial x}{\partial x_j}+\frac{\partial y}{\partial x_i}\frac{\partial y}{\partial x_j}+\frac{\partial z}{\partial x_i}\frac{\partial z}{\partial x_j}$

$G=\begin{pmatrix} g_{11} & g_{12} & g_{13} \\ g_{21} & g_{22} & g_{23} \\ g_{31} & g_{32} & g_{33} \end{pmatrix}$

柱坐标 $ds^2=[d(r\cos\phi)]^2+[d(r\sin\phi)]^2+dz^2$

$=(\cos\phi\,dr-r\sin\phi\,d\phi)^2+(\sin\phi\,dr+r\cos\phi\,d\phi)^2+dz^2$

$=dr^2+r^2d\phi^2+dz^2$

球坐标 $ds^2=dr^2+r^2d\theta^2+r^2\sin^2\theta\,d\phi^2$

无交叉项 ~~$dr\,d\phi$~~, ~~$d\theta\,d\phi$~~ ....

即 $ds^2=(h_1dx_1)^2+(h_2dx_2)^2+(h_3dx_3)^2$

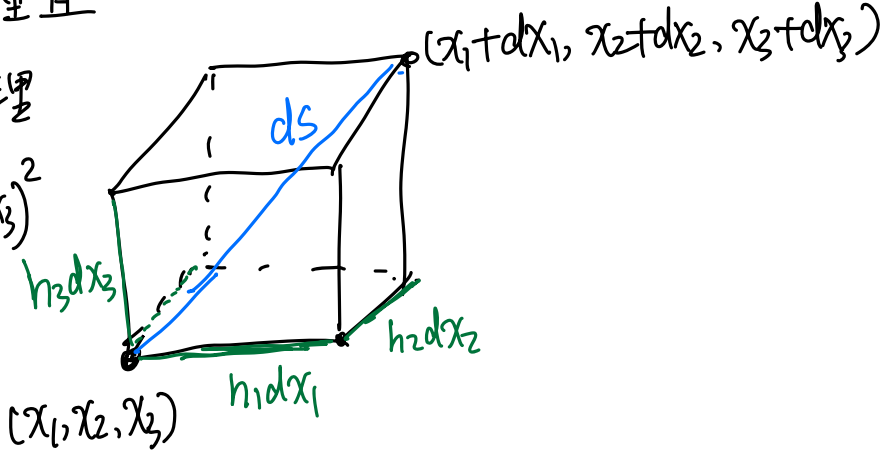
$g_{11}=(h_1)^2$ $g_{22}=(h_2)^2$

ds^2 无交叉项 $\Longrightarrow$ 正交曲线坐标系。 判据

理解：① 三条坐标线垂直

$\Updownarrow$ 勾股定理

$$ds^2=(h_1dx_1)^2+(h_2dx_2)^2+(h_3dx_3)^2$$



② $\vec{v_i}=\left(\frac{\partial x}{\partial x_i}, \frac{\partial y}{\partial x_i}, \frac{\partial z}{\partial x_i}\right)$ 沿第 i 个坐标线方向的切向量

$$g_{ij}=\vec{v_i}\cdot\vec{v_j}=0 \quad (i\neq j)$$

坐标线正交

抛物坐标 $dx=\frac{\partial x}{\partial \xi}d\xi+\frac{\partial x}{\partial \eta}d\eta+\frac{\partial x}{\partial \phi}d\phi=\cdots$，$dy=\cdots$，$dz=\cdots$

$$\Rightarrow ds^2=dx^2+dy^2+dz^2=\frac{\xi+\eta}{4\xi}(d\xi)^2+\frac{\xi+\eta}{4\eta}(d\eta)^2+\xi\eta(d\phi)^2$$

坐标 + 度规 $\longrightarrow$ 距离

坐标本身不代表距离，甚至不必是长度量纲（例如 θ,ϕ 无量纲）

微分算子

$\nabla u = ?$

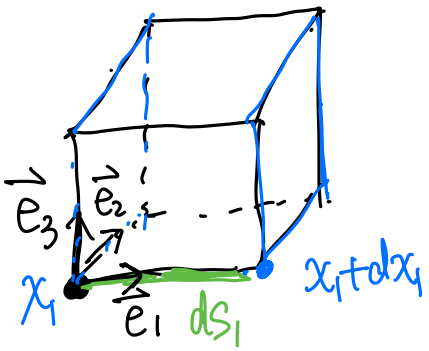
① 链式法则硬算
② 外微分，星算子
③ 直观推导

(一) 梯度 $\nabla u = ?$

直角坐标系下

$$\nabla u = \frac{\partial u}{\partial x}\vec{e}_x + \frac{\partial u}{\partial y}\vec{e}_y + \frac{\partial u}{\partial z}\vec{e}_z$$

$\frac{\partial u}{\partial x}$：x方向单位向量



∇u 沿 $\vec{e}_1$ 方向分量

$$\frac{du}{ds_1} = \frac{\partial u}{h_1 \partial x_1} = \frac{1}{h_1}\frac{\partial u}{\partial x_1}$$

$$ds^2 = (h_1 dx_1)^2 + (h_2 dx_2)^2 + (h_3 dx_3)^2$$

$h_1 dx_1 = ds_1$，$h_2 dx_2 = ds_2$，$h_3 dx_3 = ds_3$

dx_1 坐标距离
ds_1 实际距离

$$\Rightarrow \nabla u = \frac{1}{h_1}\frac{\partial u}{\partial x_1}\vec{e}_1 + \frac{1}{h_2}\frac{\partial u}{\partial x_2}\vec{e}_2 + \frac{1}{h_3}\frac{\partial u}{\partial x_3}\vec{e}_3$$

柱坐标系 $ds^2 = dr^2 + r^2 d\phi^2 + dz^2$

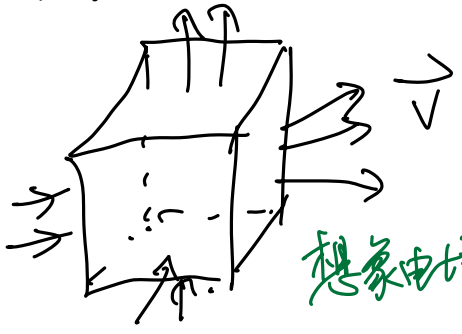
$h_r = 1 \quad h_\phi = r \quad h_z = 1$

$$\nabla u = \frac{\partial u}{\partial r}\vec{e}_r + \frac{1}{r}\frac{\partial u}{\partial \phi}\vec{e}_\phi + \frac{\partial u}{\partial z}\vec{e}_z$$

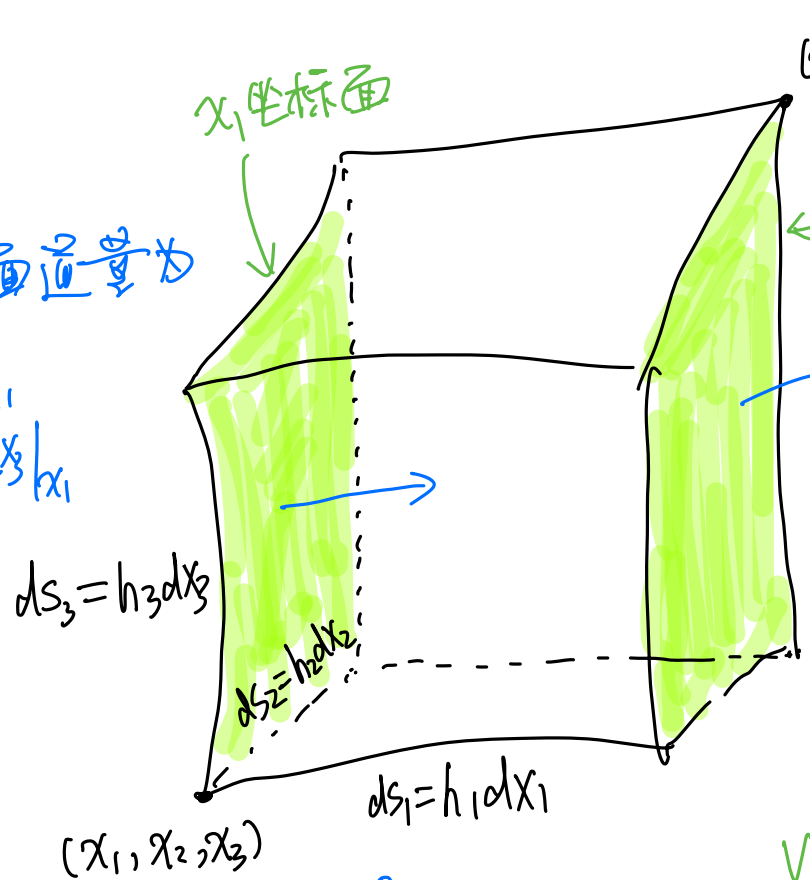
(二) 散度 $\nabla \cdot \vec{V} = ?$

直角坐标系下

$$\nabla \cdot \vec{V} = \frac{\partial V_x}{\partial x} + \frac{\partial V_y}{\partial y} + \frac{\partial V_z}{\partial z}$$



$$\nabla \cdot \vec{V} = \left(\frac{\text{穿过表面向外 } \vec{V} \text{ 的净通量}}{\text{体积}}\right)_{\text{体积}\to 0}$$



曲线坐标系中的小"立方体"，其每条棱均为坐标线

左右两面 前后两面 上下两面

$$\text{净通量}: \left[\frac{\partial(V_1 h_2 h_3)}{\partial x_1} + \frac{\partial(V_2 h_3 h_1)}{\partial x_2} + \frac{\partial(V_3 h_1 h_2)}{\partial x_3}\right] dx_1 dx_2 dx_3$$

$$\Rightarrow \nabla\cdot\vec{V} = \frac{\text{净通量}}{h_1 h_2 h_3 dx_1 dx_2 dx_3} = \frac{1}{h_1 h_2 h_3}\left[\frac{\partial}{\partial x_1}(V_1 h_2 h_3) + \frac{\partial}{\partial x_2}(V_2 h_3 h_1) + \frac{\partial}{\partial x_3}(V_3 h_1 h_2)\right]$$

电势

$$(\equiv) \text{ Laplacian } \nabla^2 u = \nabla\cdot(\nabla u)$$

直角坐标系

$$\nabla u = \frac{\partial u}{\partial x}\vec{e}_x + \frac{\partial u}{\partial y}\vec{e}_y + \frac{\partial u}{\partial z}\vec{e}_z$$

$$\nabla\cdot(\nabla u) = \frac{\partial}{\partial x}\left(\frac{\partial u}{\partial x}\right) + \frac{\partial}{\partial y}\left(\frac{\partial u}{\partial y}\right) + \frac{\partial}{\partial z}\left(\frac{\partial u}{\partial z}\right)$$

$V_i = \frac{1}{h_i}\frac{\partial u}{\partial x_i}$

来自面积

$$\nabla^2 u = \frac{1}{h_1 h_2 h_3}\left[\frac{\partial}{\partial x_1}\left(\frac{h_2 h_3}{h_1}\frac{\partial u}{\partial x_1}\right) + \frac{\partial}{\partial x_2}\left(\frac{h_3 h_1}{h_2}\frac{\partial u}{\partial x_2}\right) + \frac{\partial}{\partial x_3}\left(\frac{h_1 h_2}{h_3}\frac{\partial u}{\partial x_3}\right)\right]$$

来自体积　　梯度

每个符号的几何意义都简单明确

柱坐标 $h_r=1, h_\phi=r, h_z=1 \Rightarrow \nabla^2 u=\frac{1}{r}\left[\frac{\partial}{\partial r}\left(r\frac{\partial u}{\partial r}\right)+\frac{\partial}{\partial \phi}\left(\frac{1}{r}\frac{\partial u}{\partial \phi}\right)+\frac{\partial}{\partial z}\left(r\frac{\partial u}{\partial z}\right)\right]$

$=\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial u}{\partial r}\right)+\frac{1}{r^2}\frac{\partial^2 u}{\partial \phi^2}+\frac{\partial^2 u}{\partial z^2}$

球坐标 $ds^2=dr^2+r^2d\theta^2+r^2\sin^2\theta d\phi^2$, $h_r=1, h_\theta=r, h_\phi=r\sin\theta$

$\Rightarrow \nabla^2 u=\frac{1}{r^2\sin\theta}\left[\frac{\partial}{\partial r}\left(r^2\sin\theta\frac{\partial u}{\partial r}\right)+\frac{\partial}{\partial \theta}\left(\frac{r\sin\theta}{r}\frac{\partial u}{\partial \theta}\right)+\frac{\partial}{\partial \phi}\left(\frac{r}{r\sin\theta}\frac{\partial u}{\partial \phi}\right)\right]$

$=\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial u}{\partial r}\right)+\frac{1}{r^2\sin\theta}\frac{\partial}{\partial \theta}\left(\sin\theta\frac{\partial u}{\partial \theta}\right)+\frac{1}{r^2\sin^2\theta}\frac{\partial^2 u}{\partial \phi^2}$

高次项问题

二维空间 $\nabla^2 u=\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}$ ⟶ 曲线坐标系 $ds^2=(h_1dx_1)^2+(h_2dx_2)^2$

$\nabla^2 u=\frac{1}{h_1h_2}\left[\frac{\partial}{\partial x_1}\left(\frac{h_2}{h_1}\frac{\partial u}{\partial x_1}\right)+\frac{\partial}{\partial x_2}\left(\frac{h_1}{h_2}\frac{\partial u}{\partial x_2}\right)\right]$

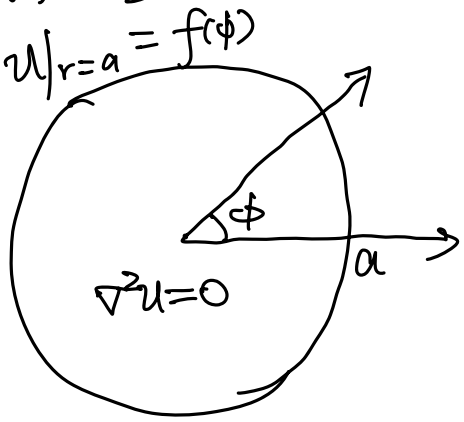
推导：将二维空间补充为三维（例如极坐标⟶柱坐标），$h_3=1$；

$u(x_1,x_2)$不依赖于x_3时，三维 Laplacian 化为二维 Laplacian.

四维空间

$\nabla^2 u=\frac{1}{h_1h_2h_3h_4}\left[\frac{\partial}{\partial x_1}\left(\frac{h_2h_3h_4}{h_1}\frac{\partial u}{\partial x_1}\right)+\frac{\partial}{\partial x_2}\left(\frac{h_3h_4h_1}{h_2}\frac{\partial u}{\partial x_2}\right)+\frac{\partial}{\partial x_3}\left(\frac{h_4h_1h_2}{h_3}\frac{\partial u}{\partial x_3}\right)+\frac{\partial}{\partial x_4}\left(\frac{h_1h_2h_3}{h_4}\frac{\partial u}{\partial x_4}\right)\right]$

分离变量



$$\nabla^2 u = 0 \qquad \underline{x^2+y^2<a^2} \quad r<a$$

$$u|_{r=a} = f(\phi)$$

$$\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial u}{\partial r}\right)+\frac{1}{r^2}\frac{\partial^2 u}{\partial \phi^2}=0, \quad \begin{matrix}0<r<a\\ 0<\phi<2\pi\end{matrix}$$

$$\begin{cases} u|_{r=a}=f(\phi) \\ u|_{\phi=0}=u|_{\phi=2\pi}, \ \left.\frac{\partial u}{\partial \phi}\right|_{\phi=0}=\left.\frac{\partial u}{\partial \phi}\right|_{\phi=2\pi} \\ u|_{r=0}<\infty \ \text{(有界条件)} \end{cases}$$

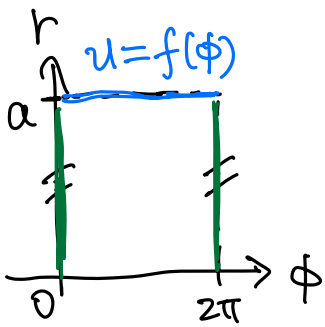
$$u=R(r)\Phi(\phi)$$

$$\Rightarrow \begin{cases} r\frac{d}{dr}\left(r\frac{dR}{dr}\right)-\lambda R=0 \\ R(0)<\infty \end{cases}$$

$$\begin{cases} \Phi''(\phi)+\lambda\Phi(\phi)=0 \\ \Phi(0)=\Phi(2\pi), \ \Phi'(0)=\Phi'(2\pi) \end{cases}$$

周期边界条件 (PBC)

↑ periodic



1) $\lambda=0, \Rightarrow \Phi_0(\phi)=1$

2) $\lambda<0$ 无解

3) $\lambda>0 \Rightarrow \Phi = A\sin\sqrt{\lambda}\phi + B\cos\sqrt{\lambda}\phi$

$$B = A\sin(\sqrt{\lambda}2\pi) + B\cos(\sqrt{\lambda}2\pi) \leftarrow \Phi(0)=\Phi(2\pi)$$

$$A = A\cos(\sqrt{\lambda}2\pi) - B\sin(\sqrt{\lambda}2\pi) \leftarrow \Phi'(0)=\Phi'(2\pi)$$

$$\begin{vmatrix} \sin(2\pi\sqrt{\lambda}) & \cos(2\pi\sqrt{\lambda})-1 \\ \cos(2\pi\sqrt{\lambda})-1 & -\sin(2\pi\sqrt{\lambda}) \end{vmatrix} = 0 \Rightarrow \cos(2\pi\sqrt{\lambda})=1$$

$$-\sin^2-(\cos-1)^2=-2+2\cos=0$$

$$\Downarrow$$

$\lambda_m = m^2, \quad m=1, 2, \cdots$

A, B任意

$$\Rightarrow \Phi_{m1}=\sin m\phi, \quad \Phi_{m2}=\cos m\phi$$

也可取其他本征函数 $\begin{cases} \Phi_{m1}=\sin m(\phi-\phi_0) \\ \Phi_{m2}=\cos m(\phi-\phi_0) \end{cases}$

简并 (degenerate): 一个本征值对应多个本征函数，取为正交以便计算级数系数

周期边界条件下常有简并

复本征函数

$\cos m\phi \pm i \sin m\phi = e^{\pm im\phi}$, $\Phi_{m1} = e^{im\phi}$, $\Phi_{m2} = e^{-im\phi}$, $m=1,2,\cdots$

$\lambda_m = m^2$, $\Phi_m = e^{im\phi}$, $m=0,\pm1,\pm2,\cdots$

$\int_0^{2\pi} \Phi_m^*(\phi)\Phi_n(\phi)d\phi = 0$, $m \neq n$

$\langle f,g\rangle = \int_0^{2\pi} f^*(\phi)g(\phi)d\phi$

$\Downarrow$

$\langle f,f\rangle$ 实数

$f(\phi) = \sum_{n=0,\pm1,\pm2\cdots} C_n e^{in\phi} \Rightarrow \int_0^{2\pi} f(\phi)(e^{in\phi})^* d\phi = 2\pi C_n$

$\Rightarrow C_n = \frac{1}{2\pi}\int_0^{2\pi} f(\phi)e^{-in\phi}d\phi$

傅里叶级数

$\frac{d}{dt}\left(r\frac{d}{dr}\left(r\frac{dR}{dr}\right)\right) - \lambda R = 0$

$\Downarrow$

$r^2R'' + rR' - \lambda R = 0$ ← 欧拉方程

$r = e^t$, $\frac{d}{dt} = \frac{dr}{dt}\frac{d}{dr} = r\frac{d}{dr}$

$\frac{d^2R(t)}{dt^2} - \lambda R(t) = 0$

1) $\lambda = 0 \Rightarrow R_0(t) = C_0 + D_0 t = C_0 + D_0 \ln r$

2) $\lambda_m = m^2$, $m = 1, 2\cdots$

$R_m = C_m e^{mt} + D_m e^{-mt}$

$= C_m r^m + D_m r^{-m}$

$\Rightarrow u(r,\phi) = C_0 + D_0 \ln r + \sum_{m=1}^{\infty}(C_{m1}r^m + D_{m1}r^{-m})\sin m\phi$

$+ \sum_{m=1}^{\infty}(C_{m2}r^m + D_{m2}r^{-m})\cos m\phi$

$u|_{r=0} < \infty$

$u|_{r=a} = f(\phi) = C_0 + \sum_{m=1}^{\infty} a^m(C_{m1}\sin m\phi + C_{m2}\cos m\phi)$

左右同乘以本征函数，由正交性得出系数：

$C_0 = \frac{1}{2\pi}\int_0^{2\pi} f(\phi)d\phi$,

$C_{m1} = \frac{1}{\pi a^m}\int_0^{2\pi} f(\phi)\sin m\phi\, d\phi$

$C_{m2} = \frac{1}{\pi a^m}\int_0^{2\pi} f(\phi)\cos m\phi\, d\phi$

$u(r,\phi) = C_0 + \sum_{m=1}^{\infty}(C_{m1}\sin m\phi + C_{m2}\cos m\phi)r^m$

可进一步化简：

$$u(r,\phi)=\frac{1}{2\pi}\int_0^{2\pi} f(\phi')d\phi' + \sum_{m=1}^{\infty}\left[\frac{1}{a^m\pi}\int_0^{2\pi} f(\phi')\sin m\phi' d\phi'\right] r^m \sin m\phi$$

$$+\sum_{m=1}^{\infty}\left[\frac{1}{a^m\pi}\int_0^{2\pi} f(\phi')\cos m\phi' d\phi'\right] r^m \cos m\phi$$

$$=\frac{1}{2\pi}\int_0^{2\pi} f(\phi')d\phi' + \sum_{m=1}^{\infty}\left(\frac{r}{a}\right)^m \frac{1}{\pi}\int_0^{2\pi} f(\phi')\cos m(\phi-\phi')d\phi'$$

$$=\frac{1}{2\pi}\int_0^{2\pi} f(\phi')\left[1+2\sum_{m=1}^{\infty}\left(\frac{r}{a}\right)^m \cos m(\phi-\phi')\right]d\phi'$$

→ $R\to\frac{r}{a}$, $\phi\to\phi-\phi'$

利用公式 $1+2\sum_{m=1}^{\infty} R^m\cos m\phi = \mathrm{Re}\left(1+2\sum_{m=1}^{\infty} R^m e^{im\phi}\right)$

$$=\mathrm{Re}\left(1+2\frac{Re^{i\phi}}{1-Re^{i\phi}}\right)=\mathrm{Re}\left(\frac{1+Re^{i\phi}}{1-Re^{i\phi}}\right)$$

$$=\mathrm{Re}\left[\frac{(1+Re^{i\phi})(1-Re^{-i\phi})}{(1-Re^{i\phi})(1-Re^{-i\phi})}\right]=\frac{1-R^2}{1+R^2-2R\cos\phi}$$

$P_R(\phi)$, Poisson 核

$(-1<R<1)$

$$=\frac{1}{2\pi}\int_0^{2\pi} f(\phi')\frac{1-\frac{r^2}{a^2}}{1+\frac{r^2}{a^2}-2\frac{r}{a}\cos(\phi-\phi')}d\phi'$$

$$=\frac{a^2-r^2}{2\pi}\int_0^{2\pi}\frac{f(\phi')}{a^2+r^2-2ar\cos(\phi-\phi')}d\phi'$$

↑

Poisson 积分公式

Helmholtz方程 $\nabla^2 u + k^2 u = 0$ $u(r,\phi,z)$

1) 柱坐标 $\frac{1}{r}\frac{\partial}{\partial r}(r\frac{\partial u}{\partial r}) + \frac{1}{r^2}\frac{\partial^2 u}{\partial \phi^2} + \frac{\partial^2 u}{\partial z^2} + k^2 u = 0$

① $u = v(r,\phi) Z(z)$

$\Rightarrow [\frac{1}{r}\frac{\partial}{\partial r}(r\frac{\partial v}{\partial r}) + \frac{1}{r^2}\frac{\partial^2 v}{\partial \phi^2} + k^2 v]\frac{1}{v} = -\frac{d^2 Z}{dz^2}\frac{1}{Z} = \lambda$

$\Rightarrow Z'' + \lambda Z = 0$

$\frac{1}{r}\frac{\partial}{\partial r}(r\frac{\partial v}{\partial r}) + \frac{1}{r^2}\frac{\partial^2 v}{\partial \phi^2} + (k^2 - \lambda) v = 0$

② $v(r,\phi) = R(r)\Phi(\phi)$ ⇓

$[\frac{1}{r}\frac{d}{dr}(r\frac{dR}{dr}) + (k^2-\lambda)R]\frac{r^2}{R} = -\frac{d^2\Phi}{d\phi^2}\frac{1}{\Phi} = \mu$

$\Rightarrow \Phi'' + \mu\Phi = 0$,

$\frac{1}{r}\frac{d}{dr}(r\frac{dR}{dr}) + (k^2 - \lambda - \frac{\mu}{r^2})R = 0$ ← Bessel型方程

Bessel eq. 标准形式

$\frac{1}{x}\frac{d}{dx}(x\frac{dR}{dx}) + (1 - \frac{\nu^2}{x^2})R = 0$

球坐标 $\frac{1}{r^2}\frac{\partial}{\partial r}(r^2\frac{\partial u}{\partial r}) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial \theta}(\sin\theta\frac{\partial u}{\partial \theta}) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2 u}{\partial \phi^2} + k^2 u = 0$

① $u(r,\theta,\phi) = R(r) S(\theta,\phi)$

$\Rightarrow [\frac{1}{r^2}\frac{d}{dr}(r^2\frac{dR}{dr}) + k^2 R]S = -\frac{1}{r^2}[\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}(\sin\theta\frac{\partial S}{\partial\theta}) + \frac{1}{\sin^2\theta}\frac{\partial^2 S}{\partial\phi^2}]R$

$\Rightarrow \frac{r^2}{R}[\frac{1}{r^2}\frac{d}{dr}(r^2\frac{dR}{dr}) + k^2 R] = -\frac{1}{S}[\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}(\sin\theta\frac{\partial S}{\partial\theta}) + \frac{1}{\sin^2\theta}\frac{\partial^2 S}{\partial\phi^2}] = \lambda$

$$\Rightarrow \frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right)+\left(k^2-\frac{\lambda}{r^2}\right)R=0 \leftarrow \text{球 Bessel 方程}$$

$$\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial S}{\partial\theta}\right)+\frac{1}{\sin^2\theta}\frac{\partial^2 S}{\partial\phi^2}+\lambda S=0$$

② $S(\theta,\phi)=\Theta(\theta)\Phi(\phi)$

$$\Rightarrow \frac{\sin^2\theta}{\Theta}\left[\frac{1}{\sin\theta}\frac{d}{d\theta}\left(\sin\theta\frac{d\Theta}{d\theta}\right)+\lambda\Theta\right]=-\frac{1}{\Phi}\frac{d^2\Phi}{d\phi^2}=\mu$$

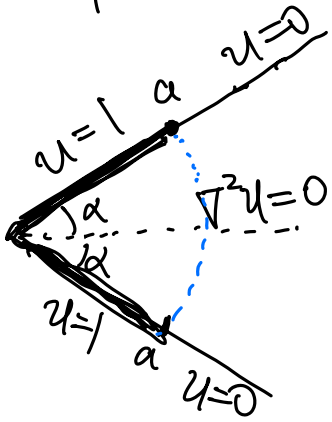
$$\Rightarrow \frac{1}{\sin\theta}\frac{d}{d\theta}\left(\sin\theta\frac{d\Theta}{d\theta}\right)+\left(\lambda-\frac{\mu}{\sin^2\theta}\right)\Theta=0$$

$$\frac{d^2\Phi}{d\phi^2}+\mu\Phi=0$$

↑ 连带 Legendre 方程

$\mu=0$ 时称为 Legendre 方程

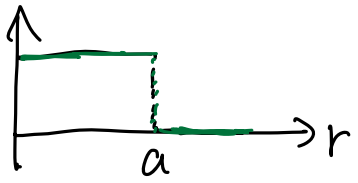
例：扇形稳定温度分布



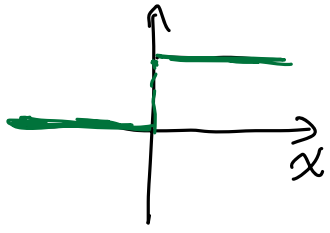
$$\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial u}{\partial r}\right)+\frac{1}{r^2}\frac{\partial^2 u}{\partial\phi^2}=0,\quad r>0,\ -\alpha<\phi<\alpha$$

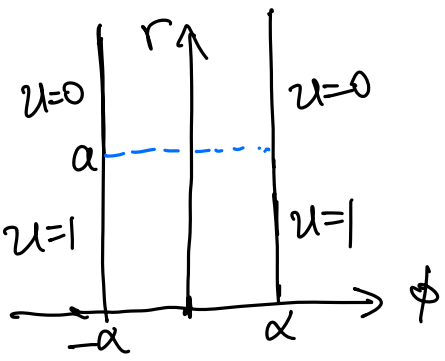
$$u|_{r=0}<\infty \qquad u|_{r\to\infty}\to 0$$

$$u|_{\phi=\pm\alpha}=\eta(a-r)=$$



单位阶跃函数 $\eta(x)=$





将扇形分割为 $r>a$ 和 $r<a$ 两区域，

连接条件：

$$u|_{r=a-0}=u|_{r=a+0},\quad \frac{\partial u}{\partial r}\Big|_{r=a-0}=\frac{\partial u}{\partial r}\Big|_{r=a+0}$$

① $r>a$ 齐次 PDE，齐次边界条件，直接分离变量法

$$u(r,\phi)=R(r)\Phi(\phi)$$

$$\int_{-\alpha}^{\alpha}\Phi_n(\phi)\Phi_m(\phi)\,d\phi=\alpha\delta_{mn}=\begin{cases}\alpha, & n=m\\ 0, & n\neq m\end{cases}$$

$$\begin{cases}\Phi_n(\phi)=\sin\frac{n\pi}{2\alpha}(\phi+\alpha), & n=1,2,3\cdots\\ \lambda_n=\left(\frac{n\pi}{2\alpha}\right)^2\end{cases}$$

↖ $\Phi_n(\pm\alpha)=0$

$$\frac{1}{r}\frac{d}{dr}\left(r\frac{dR_n}{dr}\right)-\frac{\lambda_n}{r^2}R_n=0\Rightarrow R_n=A_n\left(\frac{r}{a}\right)^{\frac{n\pi}{2\alpha}}+B_n\left(\frac{r}{a}\right)^{-\frac{n\pi}{2\alpha}}$$

$$u|_{r\to\infty}\to 0\Rightarrow A_n=0$$

$$\Rightarrow u|_{r>a}=\sum_{n=1}^{\infty}B_n\left(\frac{r}{a}\right)^{-\frac{n\pi}{2\alpha}}\sin\frac{n\pi}{2\alpha}(\phi+\alpha)$$

② $r<a$，边界条件齐次化 $u=v+w$，$v(r,\phi)=1$，解出 w

$$u|_{r<a}=v+w=1+\sum_{n=1}^{\infty}C_n\left(\frac{r}{a}\right)^{\frac{n\pi}{2\alpha}}\sin\frac{n\pi}{2\alpha}(\phi+\alpha)$$

由连接条件 $u|_{r=a-0}=u|_{r=a+0}$

$$\Downarrow$$

$$1+\sum_{n=1}^{\infty}C_n\sin\frac{n\pi}{2\alpha}(\phi+\alpha)=\sum_{n=1}^{\infty}B_n\sin\frac{n\pi}{2\alpha}(\phi+\alpha)$$

↑ $1=\sum_{n=1}^{\infty}\frac{2}{n\pi}[1-(-1)^n]\sin\frac{n\pi}{2\alpha}(\phi+\alpha)$

↖ $\frac{1}{\alpha}\int_{-\alpha}^{\alpha}1\cdot\sin\frac{n\pi}{2\alpha}(\phi+\alpha)\,d\phi$

$$\frac{\partial u}{\partial r}\Big|_{r=a-0}=\frac{\partial u}{\partial r}\Big|_{r=a+0}$$

$$\Downarrow$$

$$\sum_{n=1}^{\infty}nC_n\sin\frac{n\pi}{2\alpha}(\phi+\alpha)=-\sum_{n=1}^{\infty}nB_n\sin\frac{n\pi}{2\alpha}(\phi+\alpha)$$

$$\Downarrow$$

$$C_n=-B_n$$

$$\Downarrow$$

$$\frac{2}{n\pi}[1-(-1)^n]+C_n=B_n$$

$$\Rightarrow B_n=\frac{1}{n\pi}[1-(-1)^n]=-C_n$$

$$\Rightarrow u(r,\phi)=\begin{cases}\frac{2}{\pi}\sum_{n\,odd}\frac{1}{n}\left(\frac{r}{a}\right)^{-\frac{n\pi}{2\alpha}}\sin\frac{n\pi}{2\alpha}(\phi+\alpha) & r>a\\ 1-\frac{2}{\pi}\sum_{n\,odd}\frac{1}{n}\left(\frac{r}{a}\right)^{\frac{n\pi}{2\alpha}}\sin\frac{n\pi}{2\alpha}(\phi+\alpha) & r<a\end{cases}$$

$$\Downarrow \sin[\theta+(k+\tfrac{1}{2})\pi]=(-1)^k\cos\theta$$

$n=2k+1,\ k=0,1,2\cdots$ $\Downarrow$ 等价写法

$$u(r,\phi)=\begin{cases}\frac{2}{\pi}\sum_{k=0}^{\infty}\frac{(-1)^k}{2k+1}\left(\frac{r}{a}\right)^{-\frac{(2k+1)\pi}{2\alpha}}\cos\frac{2k+1}{2\alpha}\pi\phi & r>a\\ 1-\frac{2}{\pi}\sum_{k=0}^{\infty}\frac{(-1)^k}{2k+1}\left(\frac{r}{a}\right)^{\frac{(2k+1)\pi}{2\alpha}}\cos\frac{2k+1}{2\alpha}\pi\phi & r<a\end{cases}$$

图示：

